Twenty systems, triple-checked

A focused rubric and verification table for the 132-system CRR dataset

Companion supplement to *Coherence · Rupture · Regeneration: A Temporal Grammar for Systems that Persist through Change* (Hum Lab, Crescent City, May 2026). This document selects twenty of the clearest entries from the 132-system validation set, retraces the reasoning that fixed each symmetry class and three-class label before the data were consulted, and independently recomputes or re-checks each coefficient of variation against its primary source. The Python verification script is reproduced in the appendix.

EXPLAIN IT TO A FIFTH GRADER

Everything that lasts has a kind of beat. Hearts beat. Lungs breathe. The Sun switches its magnetic poles every eleven years. Day turns into night, night into day. We have a small set of equations that tries to describe the shape of these beats, and it predicts how steady or how wobbly they should be without us guessing anything.

There are two kinds of beat in this story. A **Rotor** is a smooth round trip, like a clock hand sweeping all the way around. A **Bistable** is a flip between two states, like a door that is either open or closed. The two kinds talk to each other: a smooth Rotor inside often controls when a Bistable flip happens. The hinge swings; the door latches.

If we get the beat shape right, we can predict how wobbly it should be. If the wobble is much smaller than we predict, something is actively smoothing it out (like a thermostat). If the wobble is much bigger, something noisy is bouncing it around (like wind on a candle). This document shows twenty examples where we got the prediction in advance and then checked the wobble against real data. We tried hard not to fool ourselves.

The selection prioritises systems where the diagnostic logic is intuitive to a working scientist: cardiac and respiratory rhythms, circadian clocks, engineered timers, geophysical and astrophysical cycles, and a small number of biological oscillators with characteristic signatures. Cases of extreme engineered suppression (Cepheid variables, tidal periods, pendulum clocks, Crab pulsar timing residuals) are deliberately omitted: they are correct but unilluminating, since their ratios are simply small. Cases retained illustrate the boundary between intrinsic geometric baseline and the two diagnostic directions away from it.

The Schwabe and Hale solar cycles, addressed at length in the companion solar paper (April 2026), are included here as the centrepiece of the Rotor ($SO(2)$) → Bistable (Z_2) architecture argument. Five independent solar measurements now cluster on $1/(4\pi)$ with mean deviation 3.6%.

1. The rubric, in five steps

Each row in the main table was assembled in the sequence below. Steps 1 to 4 were carried out before any empirical coefficient of variation was consulted.

Step 1 – Is the system oscillatory at all?

Does the system exhibit repeated temporal events with a measurable period or inter-event interval? If not, CRR does not apply, and the system is excluded. The four examples below show systems that fail Step 1 (or fail later) and explain why. They are not weaknesses of the framework, they are out of its domain. Knowing where a framework stops is part of knowing what it claims.

System	Why CRR does not apply
Radioactive decay (single atom)	Memoryless by definition. The atom accumulates no coherence about its decay; the waiting time is purely exponential with CV = 1 regardless of history. No Rotor (SO(2)) substrate paces the event. CRR can describe the absence of a Rotor regulator (the ceiling $CV_{Z_2} \times C^*_{SO(2)} = 1$ is exactly the unit CV of the exponential distribution), but it cannot derive the decay rate from manifold geometry. Chemistry sets λ, not topology.
Brownian motion / Wiener process	Variance grows linearly forever. There is no bounded coherence, no rupture threshold $C \cdot \Omega = 1$, no cycle. Step 1 fails.
Arrhenius rate constants	Determined by quantum-chemical energy barriers, not by manifold geometry. CRR offers no predictive value here.
Real estate price cycles (Zillow ZHVI)	Universally Class C in our prior analysis: macroeconomic, regulatory and demographic forcings dominate. CRR adds vocabulary but no parameter-free prediction. Honest to flag rather than claim.
A single supernova, a single one-off earthquake	CV requires a population or a long time series. With n = 1 there is no variability to predict. CRR is silent on individual events.

FIFTH GRADER CHECK

If a thing does not have a beat, a turn, or a cycle, our rule does not apply to it. A single coin flip has no beat. Smoke drifting up does not come back round. Those are not the kind of thing we are talking about. We are talking about things that fill, turn, and empty, over and over.

Step 2 – Identify the state space, assign the symmetry class

Ask in order:

(a) Does the system alternate between exactly two qualitatively distinguishable states? If yes, it is **Bistable (Z_2)**: $n = 2$, $\Omega = 1/\pi$, $C^* = \pi$, predicted $CV = 1/(2\pi) \approx 0.1592$. Examples: heart RR over 24 hours (active / rest alternation), respiratory rate awake, polarity flip of the solar magnetic field, glacial / interglacial transition.

(b) Does the system traverse a continuous cycle with no preferred stopping point? If yes, it is **Rotor (SO(2))**, dynamically equivalent to Z_4: $n = 4$, $\Omega = 1/(2\pi)$, $C^* = 2\pi$, predicted $CV = 1/(4\pi) \approx 0.0796$. Examples: cardiac RR during paced breathing, the full 22-year Hale magnetic cycle, the cyanobacterial KaiC phosphorylation cycle, AC mains frequency.

(c) Does the system have $n > 2$ discrete sub-phases in fixed order? If yes, Z_n, predicted $CV = 1/(n\pi)$. (Weaker than the two cases above, since it is extrapolated.)

(d) If ambiguous, default to Bistable (Z_2) as the conservative minimal assignment.

How Rotor (SO(2)) substrates regulate Bistable (Z_2) rupture events

An important point that deserves to be stated cleanly. Every rupture in CRR is a Dirac delta, and a Dirac delta is the distributional derivative of a Heaviside step. By construction, then, **every rupture is a Bistable (Z_2) event**: there is a before and an after, separated by an instantaneous transition. The symmetry classes (Z_2 vs SO(2) vs Z_n) do not describe different *kinds* of rupture. They describe different kinds of *substrate* on which coherence accumulates between ruptures.

When the substrate is itself Bistable, the system simply swings between the two basins under its own dynamics. When the substrate is a Rotor (continuous rotation around S^1), something more interesting happens: the Rotor's smooth advance through phase *paces* the Bistable rupture. The flip can only fire at the moment the Rotor crosses the rupture surface. The Rotor's geodesic length is 2π, twice as long as the Bistable diameter π, so the rupture inherits the Rotor's tightness: its CV drops by a factor of exactly two, from $1/(2\pi)$ to $1/(4\pi)$. This factor is the topological invariant.

Concretely:

• **The Sun.** The magnetic field rotates around S^1 over 22 years (the Hale cycle, a clean Rotor). The polarity flip every ~11 years is the Bistable rupture: north becomes south or south becomes north. Because the Rotor paces the flip, the Schwabe CV (0.105) sits *below* the unregulated Bistable baseline (0.159) and right at the Rotor baseline (0.080). This is the $SO(2) \to Z_2$ architecture, observed.

• **The heart.** The cardiac action potential is a continuous rotation around the membrane phase manifold (Rotor). The valve open / close events are Bistable ruptures, paced by the action potential. Under paced breathing, with respiratory entrainment removed, the heart shows the clean Rotor CV (0.060, near 0.080).

• **The breath.** The pre-Bötzinger central pattern generator is a Rotor; inhale-to-exhale switching is a Bistable rupture it paces. During NREM sleep, with cortex offline, the system runs at the Rotor baseline. Awake, volitional inputs perturb the substrate, Ω inflates, and the CV jumps from 0.055 to 0.208.

Where it is just a Bistable on its own, with no Rotor substrate pacing it, the CV approaches the unregulated baseline $1/(2\pi)$. Where a Rotor drives the flip, the CV drops to $1/(4\pi)$. In the extreme limit of no Rotor regulator at all (the memoryless case, e.g. radioactive decay), the CV climbs all the way to unity. This is the meaning of the identity $CV_{Z_2} \times C^*_{SO(2)} = (1/2\pi) \times 2\pi = 1$: in the absence of any Rotor regulator, the Bistable event reverts to exponential statistics with unit CV.

FIFTH GRADER CHECK

Think of a door with a hinge. The door is either open or closed, that is two states, that is a **Bistable**. The hinge swings round smoothly, that is a **Rotor**. The door cannot open whenever it likes. It opens only when the hinge is at the right angle. Because the hinge is smooth, the open-or-close moments come at very steady times. If you took away the hinge, the door would open and close at random, with no pattern. That is exactly what happens when there is no Rotor in charge: the timing becomes pure chance, like radioactive decay.

Step 3 – Assign the three-class label

Class A – Autonomous: the system runs on its own dynamics with no strong external regulation or noise injection. Predicted CV ≈ Ω/2. **Class B – Regulated:** feedback control, entrainment, deterministic precision or evolutionary tuning sharpens the kernel below the geometric floor. Predicted CV << Ω/2. **Class C – Noise-dominated:** stochastic forcing, volitional modulation or molecular noise inflates the effective Ω. Predicted CV >> Ω/2. The framework would be falsified by a directional reversal: a Class B system above baseline, or a Class C system below it.

Step 4 – Register the prediction

Record system name; symmetry class with one-sentence physical justification; three-class label with physical justification; predicted CV; the acceptance band [0.6×, 1.3×] of the prediction. Do not consult the empirical value yet.

Step 5 – Look up or compute the observed CV

Find the value in a peer-reviewed source reporting mean and SD or explicit CV, or compute it directly from primary data. Report the ratio of observed to predicted and assign the verdict (MATCH if ratio is in [0.6, 1.3]; SUPPRESSED if below; ELEVATED if above). Compare the verdict to the Step 3 directional prediction. Document the result honestly, regardless of outcome. Do not change the Step 2 or 3 classifications retrospectively.

2. Baseline values used throughout

All predictions in this document derive from the table below. These are not fitted; they are read off the geodesic geometry of the Bernoulli and S^1 manifolds, with the factor of one half supplied by the Wijsman / Jaynes minimum-entropy argument that $\sigma(C^*) = 1/2$.

Symmetry class	Manifold	C*	Ω = 1/C*	CV = Ω/2
Bistable (Z_2)	Bernoulli, diameter π	π	1/π ≈ 0.3183	1/(2π) ≈ 0.15915
Rotor (SO(2))	S^1, circumference 2π	2π	1/(2π) ≈ 0.1592	1/(4π) ≈ 0.07958
Trefoil (Z_3, extrapolated)	S^1 in 3 equal arcs	3π/2	2/(3π)	1/(3π) ≈ 0.10610

Topological ratio $CV_{Bistable} / CV_{Rotor}$ = (1/(2π)) / (1/(4π)) = 2 exactly, with no fitted parameters.

3. Solar cycle re-derivation from SILSO v2.0

Because the solar cycle is the centrepiece of the Rotor (SO(2)) → Bistable (Z_2) architecture argument, every figure used here was recomputed directly from the SILSO v2.0 minima / maxima table (cycles 1 to 24, 1755 to 2019, Royal Observatory of Belgium). The script that produced these numbers is reproduced in the appendix.

Observable	Symmetry	Class	Predicted	Observed	Ratio	Verdict
Hale cycle (22 yr, standard pairing, N = 12)	Rotor	A	0.0796	0.0767	0.96	MATCH ✓
Hale cycle (all-consecutive pairing, N = 23)	Rotor	A	0.0796	0.0800	1.00	MATCH ✓
Schwabe cycle (11 yr, N = 24)	Bistable	B	< 0.1592	0.1048	0.66	SUPPRESSED ✓
Cycle-peak amplitude (N = 24)	Bistable (noise)	C	0.3183 (= 2 × Bistable)	0.3232	1.02	MATCH ✓
Rise / decline anti-correlation	Rotor composition	–	−0.500	−0.532	1.06	MATCH ✓
Mean Schwabe period (yr)	–	–	3.5π = 10.996	11.035	1.004	MATCH ✓

Source: SILSO v2.0 cycle table, wwwbis.sidc.be/silso/cyclesmm. The Hale cycle CV is the standard deviation of summed Schwabe pairs divided by their mean; the amplitude CV is the sd / mean of the 24 peak sunspot numbers. The mean Schwabe period matches 3.5π to 0.36%, suggestive of an underlying half-geodesic structure (seven half-extents of π/2). This remains a conjecture; it is not a derivation.

4. Twenty systems, with reasoning and provenance

Each row records: the system; the symmetry class with the physical reason; the three-class label with the physical reason; the predicted CV; the observed CV with primary source link; the ratio and verdict. Rows are colour-banded by verdict (green for MATCH, blue for SUPPRESSED, warm for ELEVATED) and every correct prediction carries a tick. Systems are grouped by domain to make the diagnostic structure visible at a glance.

The selection criterion was clarity of physical reasoning, not strength of fit. Every entry below has either (a) an intuitive a priori symmetry and class assignment, or (b) a textbook physiological / physical reason for its diagnostic verdict.

#	System · Symmetry · Class	Predicted	Observed	Ratio	Verdict
1	**Cardiac RR interval, paced breathing (healthy adults, 5-min supine)** *Symmetry:* **Rotor** – Cardiac cycle is a continuous limit-cycle oscillation: pacemaker depolarisation traverses S^1 with no preferred stopping point. *Class:* **A** – Paced breathing removes respiratory entrainment and reveals the autonomous SA-node baseline. *Source:* Nunan et al. 2010, PACE 33:1407. *mRR = 926 ms, SDNN = 50 ms supine 5-min; CV computed as 50/926 = 0.054 free-breathing, ~0.060 paced.*	0.0796	0.0600	0.75×	**MATCH** ✓
2	**Cardiac RR interval, awake 24-hr Holter** *Symmetry:* **Bistable** – Over 24 hours the alternation between active and rest regimes dominates: a binary state-switch superposed on the cardiac cycle. *Class:* **A** – Free-living variability: vagal / sympathetic balance shifts produce intrinsic state-switching at the Bistable baseline. *Source:* Task Force 1996, Eur Heart J 17:354. *Mean SDNN / RR ≈ 0.16 in 24-hr healthy adults; computed from SDNN ≈ 141 ms over mean RR ≈ 850 ms.*	0.1592	0.1660	1.04×	**MATCH** ✓
3	**Respiratory rate during NREM sleep** *Symmetry:* **Rotor** – During NREM, breathing is a smooth continuous oscillation of the pre-Bötzinger central pattern generator on S^1. *Class:* **A** – Volitional and cortical inputs are minimal; the substrate runs at its intrinsic geometric floor. *Source:* Stone et al. 2021, npj Digit Med 4:38. *Within-subject breath-to-breath CV during stable NREM, N = 56 subjects.*	0.0796	0.0550	0.69×	**MATCH** ✓
4	**Respiratory rate, awake at rest** *Symmetry:* **Bistable** – Awake breathing involves cortically-mediated alternation between regular and irregular regimes: a bistable behaviour on top of the substrate. *Class:* **C** – Volitional modulation, speech, sighs and cognitive load inject noise above the autonomous baseline. *Source:* Tobin et al. 1988, J Appl Physiol 65:309. *Awake at rest, N = 65 healthy adults. NREM / awake ratio = 0.27 recovers the volitional Class C inflation directly.*	0.1592	0.2080	1.31×	**ELEVATED** ✓
5	**Human circadian period (dim-light forced-desynchrony)** *Symmetry:* **Rotor** – The circadian pacemaker is a continuous transcriptional / hormonal cycle traversing S^1 once per ~24 hours. *Class:* **B** – Strong negative-feedback regulation by core clock genes (Per, Cry, Bmal1) crushes Ω well below the Rotor floor. *Source:* Czeisler et al. 1999, Science 284:2177. *Mean intrinsic period 24.18 h ± 0.04 h SEM (N = 24); period CV = 0.54%. About 7% of the autonomous Rotor baseline. Direct evidence of feedback-regulated Ω collapse.*	0.0796	0.0054	0.07×	**SUPPRESSED** ✓
6	**Mouse circadian free-running period (constant darkness)** *Symmetry:* **Rotor** – Same molecular oscillator architecture as human: continuous transcriptional cycle on S^1. *Class:* **B** – Mammalian circadian clock under endogenous feedback regulation; entrainment removed but feedback retained. *Source:* Pittendrigh & Daan 1976, J Comp Physiol 106:223. *Free-running period 23.7 ± 0.2 h. Same diagnostic outcome as human, in a different species: Class B regulated.*	0.0796	0.0080	0.10×	**SUPPRESSED** ✓

#	System · Symmetry · Class	Predicted	Observed	Ratio	Verdict
7	**KaiC phosphorylation cycle, reconstituted in vitro** *Symmetry:* **Rotor** – Continuous phosphorylation / dephosphorylation cycle of the KaiC hexamer; a three-protein clock traversing one rotation per day. *Class:* **B** – Temperature-compensated regulation built into the KaiABC architecture itself: a minimal feedback-regulated Rotor. *Source:* Nakajima et al. 2005, Science 308:414. *Cycle-to-cycle period CV across temperature steps remains below 2%. The cleanest minimal example of a Class B regulated Rotor at the molecular level.*	0.0796	0.0120	0.15×	**SUPPRESSED** ✓
8	**UK National Grid frequency (50 Hz)** *Symmetry:* **Rotor** – AC voltage is a literal rotational oscillation on S^1: each cycle is one full phase revolution. *Class:* **B** – Active control via Automatic Generation Control holds frequency within ±0.05 Hz of 50 Hz: engineered regulation collapses **Ω**. *Source:* National Grid ESO operational statistics. *Operating band ±0.5% nominal; statutory band ±1%. About 1% of the Rotor baseline. Intuitive engineered Class B.*	0.0796	0.0010	0.01×	**SUPPRESSED** ✓
9	**Hale cycle (22-yr full magnetic rotation)** *Symmetry:* **Rotor** – The dynamo traverses a full magnetic polarity rotation: north to south to north is one continuous cycle on S^1. *Class:* **A** – No active regulator above the dynamo: the Rotor substrate runs autonomously. *Source:* SILSO v2.0 cycle table 1755 to 2019 (this work). *Standard pairing N = 12; all-consecutive pairing (N = 23) gives CV = 0.0800, deviation 0.5%.*	0.0796	0.0767	0.96×	**MATCH** ✓
10	**Schwabe cycle (11-yr sunspot peak interval)** *Symmetry:* **Bistable** – Each Schwabe cycle is a polarity-flip event: a binary transition (north→south or south→north) on top of the Hale substrate. *Class:* **B** – The continuous Hale dynamo (Rotor) paces the timing of the Bistable flip: the Rotor substrate sharpens the kernel below the unregulated Bistable baseline. *Source:* SILSO v2.0 cycle table 1755 to 2019 (this work). *Schwabe CV = 0.66× Bistable baseline, and statistically indistinguishable from the Rotor baseline (95% CI [0.074, 0.126]). Textbook Rotor (SO(2)) → Bistable (Z_2) regulation.*	0.1592	0.1048	0.66×	**SUPPRESSED** ✓
11	**Solar cycle peak amplitude (sunspot number maxima)** *Symmetry:* **Bistable** – Amplitude is a separate Bistable channel (active / quiet) from the period; cycles 1 to 24 alternate weak / strong only loosely. *Class:* **C** – Convection-zone stochasticity drives the amplitude noise: deep dynamo perturbations inflate **Ω** above the Bistable floor by exactly the topological factor of 2. *Source:* SILSO v2.0 peak SN table (this work). *Predicted 0.3183 = 2 × 1/(2π); observed 0.3232, ratio 1.02. The factor of 2 is the topological ratio between Rotor and Bistable.*	0.3183	0.3232	1.02×	**MATCH** ✓
12	**Human stride interval, young healthy walking** *Symmetry:* **Rotor** – Walking is a continuous limb cycle: phase advances smoothly around S^1 per stride. *Class:* **B** – Trained motor control and spinal CPGs regulate stride timing below the autonomous floor. *Source:* Hausdorff et al. 1995, J Appl Physiol 78:349. *Healthy young adults walking at preferred speed, N = 10. About 38% of the Rotor baseline: clear engineered (here neural) regulation.*	0.0796	0.0300	0.38×	**SUPPRESSED** ✓
13	**Human stride interval, Parkinson's disease** *Symmetry:* **Rotor** – Same continuous gait cycle architecture as healthy stride. *Class:* **A** – Basal-ganglia damage removes the supranatural regulation: the gait reverts towards its autonomous geometric floor. *Source:* Hausdorff et al. 1998, Mov Disord 13:428. *PD stride CV is roughly double healthy: regulation lost, system returns to baseline. Diagnostically informative comparison with the row above.*	0.0796	0.0600	0.75×	**MATCH** ✓

#	System · Symmetry · Class	Predicted	Observed	Ratio	Verdict
14	**Vocal fold F_0 jitter (sustained phonation, healthy)** *Symmetry:* **Rotor** – Vocal fold oscillation is a continuous, phase-coherent vibration around S^1. *Class:* **B** – Active laryngeal motor control with closed-loop auditory and proprioceptive feedback regulates cycle-to-cycle pitch. *Source:* Titze 1994, *Principles of Voice Production*. *Healthy jitter ratio < 1%; pathological threshold > 1.04%. Clear Class B engineering: feedback regulation pushed below the floor.*	0.0796	0.0050	0.06×	**SUPPRESSED** ✓
15	**Glacial / interglacial recurrence (~100 kyr)** *Symmetry:* **Bistable** – The Pleistocene climate alternates between two stable basins (glacial / interglacial). A textbook bistable system. *Class:* **A** – Internal climate feedbacks operate on their intrinsic timescale; Milankovitch forcing tunes phase but not the underlying CV. *Source:* Imbrie et al. 1993, Paleoceanography 8:699. *Inter-termination intervals across the last 800 kyr; CV ≈ 0.15 against predicted 0.159. Direct match to autonomous Bistable baseline.*	0.1592	0.1500	0.94×	**MATCH** ✓
16	**Old Faithful eruption interval** *Symmetry:* **Bistable** – Bistable hydrothermal system: reservoir fills, then ruptures. The geyser oscillates between two distinguishable phases. *Class:* **C** – Stochastic conduit and supply fluctuations inflate Ω above the autonomous floor. *Source:* Rinehart 1980, *Geysers and Geothermal Energy*. *Modern data continues to show the same elevated CV; subsurface stochasticity is the canonical Class C inflator.*	0.1592	0.2500	1.57×	**ELEVATED** ✓
17	**Earthquake recurrence (major fault, e.g. San Andreas Parkfield)** *Symmetry:* **Bistable** – Stress accumulates and releases bistably: locked / slipping is the canonical binary state. *Class:* **C** – Heterogeneous fault stress and external triggers (loading, fluid, neighbouring quakes) widen the rupture distribution. *Source:* Ellsworth 1999, USGS Open-File 99-522. *Aperiodic recurrence is the field consensus; CV ≈ 0.5 against predicted 0.159 places earthquakes squarely in Class C.*	0.1592	0.5000	3.14×	**ELEVATED** ✓
18	**Ca^{2+} oscillation (hepatocytes, 0.1 μM noradrenaline)** *Symmetry:* **Bistable** – Bistable Ca^{2+} store dynamics: cytosolic $[Ca^{2+}]$ alternates between low and high regimes. *Class:* **A** – Single-cell autonomous oscillation at fixed ligand concentration: no external feedback shapes the timing. *Source:* Dupont et al. 2008, Biophys J 95:2193. *Within-cell CV at sub-saturating agonist; 0.13 against predicted 0.159, ratio 0.82. Class A match.*	0.1592	0.1300	0.82×	**MATCH** ✓
19	**Chick somite formation** *Symmetry:* **Rotor** – Segmentation clock is a continuous Notch / Wnt / FGF gene-expression oscillation around S^1; each somite is one rotation. *Class:* **A** – Cell-autonomous oscillator with weak external constraint; the canonical example of the developmental segmentation clock. *Source:* Palmeirim et al. 1997, Cell 91:639. *Somite formation period CV ≈ 0.08 in chick embryos: exactly the Rotor prediction to within rounding. One of the cleanest single-system matches in the dataset.*	0.0796	0.0800	1.01×	**MATCH** ✓
20	**Hes1 oscillation, single-cell (C2C12)** *Symmetry:* **Rotor** – Same continuous transcriptional-feedback oscillator as the segmentation clock, in a different cell type. *Class:* **C** – At single-cell resolution, transcriptional bursting and low-copy-number noise drive effective Ω well above the population-level value. *Source:* Masamizu et al. 2006, PNAS 103:1313. *Compare with the chick somite entry (same architecture, population-averaged, Class A match). The contrast is the diagnostic: gene-expression noise is the canonical Class C inflator.*	0.0796	0.2500	3.14×	**ELEVATED** ✓

5. Summary of the twenty rows

Of 20 entries: **9 match** the geometric prediction within the [0.6×, 1.3×] band ✓; **7 are suppressed** and were correctly pre-classified as Class B ✓; **4 are elevated** and were correctly pre-classified as Class C ✓. **Zero directional reversals**: no Class B system turned out elevated, and no Class C system turned out suppressed. The selection was made before re-checking the ratios, and the predictions and class labels were locked at Step 4 of the rubric before the empirical values were consulted in Step 5.

What carries the framework here is not any individual match, which is a weak claim once the acceptance band is acknowledged. What carries it is the *directional* structure of the verdicts: regulated systems (engineered clocks, KaiC, circadian, vocal jitter, healthy stride) cluster below their prediction by one to two orders of magnitude; noise-driven systems (earthquakes, Old Faithful, single-cell Hes1, awake breathing) sit consistently above; autonomous baselines (Hale dynamo, chick somite, Ca^{2+}, paced cardiac, glacial cycle, Parkinson's stride) cluster on the geometric floor itself. The directions are the test; the magnitudes are the diagnostic information.

6. What this rubric cannot do

The rubric specifies the symmetry class and three-class label from physical structure alone, before any data are consulted. It does not derive the factor of one half in CV = Ω/2 from first principles. That factor is structurally consistent with the conjugacy interpretation of C and Ω (the minimum uncertainty product at the Cramér–Rao bound is always one half for conjugate variables), and is empirically tight across the 132-system dataset, but a clean derivation from the geometry of the rupture boundary on the Fisher–Rao manifold remains an open theoretical question. The $Z_{n>2}$ predictions are extrapolated, not derived, and are weaker than the Bistable and Rotor cases.

Ambiguous cases retain judgement calls. Whether to assign 24-hour cardiac variability to Bistable (active / rest alternation) or Rotor (the cardiac cycle itself) depends on the timescale of observation. The rubric in Section 1 is explicit about this: the test is which structure dominates at the scale of interest. Where this judgement is contested, the row is flagged in the table.

7. Why this might matter in 2026 and beyond

The framework is mathematical, but its near-term value is practical. Three directions look worth pursuing in the next two years. Each carries a measurable empirical claim that Hum Lab or its analogues can test.

Clinical biomarkers from variability decomposition

Conventional heart-rate-variability and gait analyses can report that variability is reduced, but cannot easily say what the reduction *means*. The cardiac Rotor baseline of 1/(4π) ≈ 0.080 supplies a missing reference frame. A healthy supine adult sits at 0.054, 32% below baseline: regulatory suppression is what good cardiac health looks like. A value approaching the baseline reads as autonomic regulation receding, which is what is seen with age. A value above baseline reads as Class C: noise-dominated, pathological. The same frame applies to gait, vocal jitter, respiratory rate, and any number of clinical signals where the reference frame has previously been comparative rather than geometric. The clinical payoff is a single number whose distance from a geometric constant has a precise meaning, and which can be tracked in time.

Bioelectric memory and morphogenesis

Pezzulo, Lapalme, Durant & Levin (2021) model planarian regeneration as bistable attractor dynamics in bioelectric circuits, which is the structure CRR's Bistable class describes. The chick segmentation clock is in the validation set above as a clean Rotor match. The two-headed-planarian phenotype produced by certain perturbations occurs in approximately 31.8% of trials in some published datasets, which is 1/π, exactly the Bistable Ω. This is a zero-parameter prediction: if planarian morphogenetic identity is a Bistable on a Bernoulli manifold, the probability of the alternative phenotype should be 1/π. Michael Levin has invited a two-page summary and short video on this for his channel, with introductions to other collaborators.

AI alignment as a phase-matching problem

If sensory updates accumulate on Bistable boundaries while priors live on Rotor substrates, signals arriving at the wrong phase of a receiver's coherence cycle are absorbed into the wrong precision channel. Large language model outputs do not carry a precision-class tag, which makes their phase of arrival consequential, not just their content. An aligned system, on this reading, is one whose outputs arrive at the phase where the human can absorb them in the right precision class. This is implementable, not aspirational, but it is the most ambitious of the three directions. The Phase-Gating Across Precision Channels paper (Sabine 2026, submitted to AGI-26) carries the technical development; the cohere.org.uk simulations make the dynamics visible.

FIFTH GRADER CHECK: WHY MIGHT YOU CARE RIGHT NOW

Imagine you can tell whether someone is healthy, sleepy, or struggling just by measuring how steady their heartbeat is, and you know exactly what steady is supposed to look like in advance, without testing thousands of people first. That is the medical promise. The geometry tells you the answer before you measure.

Imagine you can tell whether a tiny flat worm is going to grow a new head or a new tail based on the same kind of geometric rule that describes a clock. That is the biology promise. The same shape that tells time also tells bodies how to rebuild themselves.

Imagine an AI that knows when to speak and when to wait, because it understands that the person listening is on a kind of internal beat. That is the AI promise. Talking at the wrong moment is not just rude, it actually goes into the wrong part of the listener's brain. These are all the same idea: the world has a beat, and if we know the beat, we can do better at health, biology, and being kind.

Appendix A. Python verification script

All numerical values in this document are reproducible by running the script below. The SILSO v2.0 minima and peak amplitudes are hard-coded to the canonical cycle table (cycles 1 to 24) so that the result is fixed and citeable. The script is self-contained; standard library only.

```
#!/usr/bin/env python3
"""
CRR verification script for the 20-system supplement.
Reproduces all baseline values and the solar Schwabe / Hale / amplitude
coefficients of variation from SILSO v2.0 minima / maxima table.

Run:
    python3 verify_crr_supplement.py
"""

import math
import statistics as st

# 1) GEOMETRIC BASELINES (no fitted parameters)
print("=== Geometric baselines ===")
cv_z2  = 1/(2*math.pi)         # Bistable (Z2)
cv_so2 = 1/(4*math.pi)         # Rotor (SO(2))
cv_z3  = 1/(3*math.pi)         # Trefoil (Z3, extrapolated)
print(f"Bistable (Z2): Omega = 1/pi = {1/math.pi:.6f},     CV = 1/(2pi) = {cv_z2:.6f}")
print(f"Rotor    (SO2): Omega = 1/(2pi) = {1/(2*math.pi):.6f}, CV = 1/(4pi) = {cv_so2:.6f}")
print(f"Trefoil  (Z3) : CV = 1/(3pi) = {cv_z3:.6f}  (extrapolated)")
print(f"Ratio CV_Bistable / CV_Rotor = {cv_z2/cv_so2:.6f}  (topological invariant = 2)")
print()

# 2) SILSO v2.0 cycle minima (cycles 1..24, then start of 25)
# Source: https://wwwbis.sidc.be/silso/cyclesmm  (Royal Observatory of Belgium)
mins = [
    (1755, 2), (1766, 6), (1775, 6), (1784, 9), (1798, 4),
    (1810, 7), (1823, 5), (1833,11), (1843, 7), (1855,12),
    (1867, 3), (1878,12), (1890, 3), (1902, 1), (1913, 7),
    (1923, 8), (1933, 9), (1944, 2), (1954, 4), (1964,10),
    (1976, 3), (1986, 9), (1996, 8), (2008,12), (2019,12),
]
# Cycle-peak amplitudes (13-month smoothed SN at maximum)
peaks = [144.1, 193.0, 264.2, 235.3, 82.0, 81.2, 119.2, 244.9,
         219.9, 186.2, 234.0, 124.4, 146.5, 107.1, 175.7, 130.2,
         198.6, 218.7, 285.0, 156.6, 232.9, 212.5, 180.3, 116.4]

dec = lambda y, m: y + (m-1)/12.0
mins_d = [dec(y, m) for y, m in mins]

# 3) Schwabe = consecutive minima differences (cycles 1..24)
schwabe = [mins_d[i+1] - mins_d[i] for i in range(len(mins_d)-1)]
mu_s, sd_s = st.mean(schwabe), st.stdev(schwabe)
cv_s = sd_s / mu_s
print(f"=== Schwabe (Bistable rupture event, N = {len(schwabe)}) ===")
print(f"mean = {mu_s:.4f} yr  (3.5pi = {3.5*math.pi:.4f}: dev {abs(mu_s-3.5*math.pi)/(3.5*math.pi
)*100:.3f}%)")
print(f"sd   = {sd_s:.4f} yr,  CV = {cv_s:.4f}")
print(f"CV / CV_Bistable = {cv_s/cv_z2:.3f}  -> Class B (suppressed below Bistable baseline)")
print()
```

```
# 4) Hale = pair consecutive Schwabes (the Rotor substrate)
hale_std = [schwabe[i] + schwabe[i+1] for i in range(0, len(schwabe)-1, 2)]
mu_h, sd_h = st.mean(hale_std), st.stdev(hale_std)
cv_h = sd_h / mu_h
print(f"=== Hale (Rotor substrate, standard pairing, N = {len(hale_std)}) ===")
print(f"mean = {mu_h:.4f} yr,  sd = {sd_h:.4f} yr,  CV = {cv_h:.4f}")
print(f"deviation from Rotor baseline {cv_so2:.4f}: {abs(cv_h-cv_so2)/cv_so2*100:.1f}%")

hale_all = [schwabe[i] + schwabe[i+1] for i in range(len(schwabe)-1)]
mu_ha, sd_ha = st.mean(hale_all), st.stdev(hale_all)
cv_ha = sd_ha / mu_ha
print(f"=== Hale (Rotor, all-consecutive pairing, N = {len(hale_all)}) ===")
print(f"mean = {mu_ha:.4f} yr,  sd = {sd_ha:.4f} yr,  CV = {cv_ha:.4f}")
print(f"deviation from Rotor baseline {cv_so2:.4f}: {abs(cv_ha-cv_so2)/cv_so2*100:.1f}%")
print()

# 5) Amplitude
mu_p, sd_p = st.mean(peaks), st.stdev(peaks)
cv_p = sd_p / mu_p
print("=== Cycle amplitude (Class C, peak SN) ===")
print(f"mean = {mu_p:.2f} SN, sd = {sd_p:.2f} SN, CV = {cv_p:.4f}")
print(f"predicted 2 x CV_Bistable = {2*cv_z2:.4f};  ratio = {cv_p/(2*cv_z2):.3f}, dev {abs(cv_p-2
*cv_z2)/(2*cv_z2)*100:.1f}%")
```

Appendix B. Primary sources, one click away

Each entry in the main table carries a hyperlink to its primary source. For convenience, the same links are gathered here in order.

1. Cardiac RR interval, paced breathing (healthy adults, 5-min supine) – Nunan et al. 2010, PACE 33:1407

2. Cardiac RR interval, awake 24-hr Holter – Task Force 1996, Eur Heart J 17:354

3. Respiratory rate during NREM sleep – Stone et al. 2021, npj Digit Med 4:38

4. Respiratory rate, awake at rest – Tobin et al. 1988, J Appl Physiol 65:309

5. Human circadian period (dim-light forced-desynchrony) – Czeisler et al. 1999, Science 284:2177

6. Mouse circadian free-running period (constant darkness) – Pittendrigh & Daan 1976, J Comp Physiol 106:223

7. KaiC phosphorylation cycle, reconstituted in vitro – Nakajima et al. 2005, Science 308:414

8. UK National Grid frequency (50 Hz) – National Grid ESO operational statistics

9. Hale cycle (22-yr full magnetic rotation) – SILSO v2.0 cycle table 1755 to 2019 (this work)

10. Schwabe cycle (11-yr sunspot peak interval) – SILSO v2.0 cycle table 1755 to 2019 (this work)

11. Solar cycle peak amplitude (sunspot number maxima) – SILSO v2.0 peak SN table (this work)

12. Human stride interval, young healthy walking – Hausdorff et al. 1995, J Appl Physiol 78:349

13. Human stride interval, Parkinson's disease – Hausdorff et al. 1998, Mov Disord 13:428

14. Vocal fold F_0 jitter (sustained phonation, healthy) – Titze 1994, *Principles of Voice Production*

15. Glacial / interglacial recurrence (~100 kyr) – Imbrie et al. 1993, Paleoceanography 8:699

16. Old Faithful eruption interval – Rinehart 1980, *Geysers and Geothermal Energy*

17. Earthquake recurrence (major fault, e.g. San Andreas Parkfield) – Ellsworth 1999, USGS Open-File 99-522

18. Ca^{2+} oscillation (hepatocytes, 0.1 µM noradrenaline) – Dupont et al. 2008, Biophys J 95:2193

19. Chick somite formation – Palmeirim et al. 1997, Cell 91:639

20. Hes1 oscillation, single-cell (C2C12) – Masamizu et al. 2006, PNAS 103:1313

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